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Reinforcement Learning for Average Reward Zero-Sum Games

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ABSTRACT

We describe two Reinforcement Learning algorithms for average reward zero-sum stochastic games. One of the algorithms is based on relative Q-learning and the other on Q-learning for stochastic shortest path games. We analyze their properties and prove convergence using the ODE (Ordinary Differential Equation) method for the analysis of stochastic approximation algorithms. We further discuss the case where not all the actions are played by the opponent with comparable frequencies and suggest a method to overcome this problem.

Keywords: Q-learning, stochastic games, average reward, Reinforcement Learning.